

Regression Modelling of Air Quality Using Global Meteorological Data

Rahul Sanju
MSc. In Computing (Majors in AI)
Dublin City University
Dublin,Ireland
rahul.sanju2@mail.dcu.ie

Rahul Satheesan Nair
MSc. In Computing (Majors in AI)
Dublin City University
Dublin,Ireland
rahulsatheesan.nair2@mail.dcu.ie

Sooryakiran Vinod
MSc. In Computing (Majors in AI)
Dublin City University
Dublin,Ireland
sooryakiran.vinod2@mail.dcu.ie

Nitin Sunil Thomas
MSc. In Computing (Majors in AI)
Dublin City University
Dublin,Ireland
nitinsunil.thomas2@mail.dcu.ie

Abstract— Air pollution poses a growing threat to public health and environmental sustainability worldwide, with fine particulate matter (PM_{2.5}) recognised as one of the most harmful pollutants due to its deep penetration into the respiratory system. Accurate prediction of PM_{2.5} concentrations is critical for enabling timely interventions and informing environmental policy. Unlike the Air Quality Index, which categorises pollution into discrete bands and sacrifices predictive resolution, PM_{2.5} concentration offers a continuous and more precise representation of real-world air quality dynamics. This study presents a data mining approach to PM_{2.5} prediction, structured around the CRISP-DM framework. A comprehensive preprocessing pipeline was applied, incorporating outlier detection, log transformation, and feature interaction engineering to improve data quality and model robustness. Five machine learning models were developed and evaluated: Linear Regression, Ridge Regression, Random Forest, Support Vector Regression, and XGBoost. Models were assessed using RMSE and R^2 metrics to ensure rigorous comparison. Results show that ensemble-based non-linear models substantially outperform linear baselines, with XGBoost achieving the strongest predictive performance. Notably, experiments using meteorological features alone revealed that such variables are insufficient to fully account for PM_{2.5} variability, underscoring the influence of emission-source factors beyond meteorological conditions. These findings highlight the critical importance of thoughtful feature engineering and model selection in environmental data mining tasks, and provide a reproducible pipeline applicable to urban air quality monitoring.

Keywords— PM_{2.5} Prediction, Air Quality Monitoring, XGBoost, CRISP-DM, Feature Engineering, Meteorological Data.

I. INTRODUCTION

Air pollution is one of the most pressing environmental issues worldwide, contributing to severe health risks such as respiratory diseases, cardiovascular conditions, and reduced life expectancy [1]. Among various pollutants, fine particulate matter (PM_{2.5}) particles with a diameter of 2.5 micrometres or less is particularly hazardous due to its ability to penetrate deep into the lungs and enter the bloodstream, causing both short-term and long-term health complications. Exposure to elevated PM_{2.5} levels has been linked to increased rates of asthma, lung cancer, stroke, and premature mortality [2]. As a result, accurate prediction of

PM_{2.5} levels has become a crucial task for environmental monitoring systems, public health authorities, and smart city applications.

Traditionally, air quality has been communicated using the Air Quality Index (AQI), which categorizes pollution levels into discrete classes such as "Good," "Moderate," or "Unhealthy." While AQI is useful for public communication, it introduces a loss of information by converting continuous pollution measurements into coarse categories [21]. This discretization limits the ability of machine learning models to capture subtle variations and patterns in pollution data. In contrast, PM_{2.5} provides a continuous and more informative representation, enabling more precise modelling and prediction. For this reason, this study focuses on PM_{2.5} concentration as the primary target variable rather than AQI classification. In recent years, rapid urbanization, industrialization, and increased vehicular emissions have contributed to more complex and dynamic air pollution patterns across both developed and developing regions. These changes make traditional statistical modelling approaches, such as linear regression or time-series methods, insufficient for capturing the nonlinear and multi-dimensional relationships between environmental variables and pollutant concentrations [6]. Consequently, there is a growing need for data-driven approaches that can leverage large-scale environmental datasets to improve prediction accuracy, support real-time decision-making, and assist policymakers in designing effective air quality interventions [22].

Machine learning techniques have emerged as powerful tools for modelling such complex systems due to their ability to learn nonlinear relationships and high-order interactions among variables without requiring explicit domain-driven model specification [17]. Methods such as Random Forest, Support Vector Regression, and gradient boosting algorithms like XGBoost have demonstrated strong performance across a variety of environmental prediction tasks [5], [8]. However, the effectiveness of these models is highly dependent on data quality, preprocessing strategies, and feature engineering. Challenges such as missing values, extreme outliers, skewed distributions, and multicollinearity among meteorological variables can significantly degrade model performance if not properly addressed. Therefore, a structured and systematic data mining approach is essential to ensure reliable, reproducible, and interpretable results. The objective of this research is to develop and evaluate machine learning models

for predicting PM2.5 concentrations using meteorological and environmental data sourced from a large-scale global weather repository.

The study applies the CRISP-DM (Cross-Industry Standard Process for Data Mining) framework to guide the end-to-end analysis, covering key stages including data understanding, preprocessing, feature engineering, modelling, and evaluation. Special emphasis is placed on addressing data quality challenges, applying log transformation to handle target skewness, performing multicollinearity analysis through Variance Inflation Factor (VIF) computation, and constructing meaningful interaction features to enhance predictive performance.

This study is guided by the following research questions:

RQ1: How effectively can machine learning models predict PM2.5 concentrations using meteorological and environmental features?

RQ2: To what extent can meteorological variables alone explain PM2.5 variability, and what does this reveal about the role of emission-source factors?

To address these questions, the main objectives of this study are: (i) to develop and evaluate machine learning models for predicting PM2.5 concentrations using meteorological and environmental data, (ii) to investigate the contribution of meteorological variables in explaining variations in air quality, (iii) to apply feature engineering techniques including interaction terms, log transformation, and VIF-based selection to improve model performance, and (iv) to compare different classes of models including linear, regularized, and ensemble-based approaches in terms of predictive accuracy and interpretability.

The remainder of this paper first reviews related work in air quality prediction and machine learning for environmental modelling, followed by a description of the CRISP-DM-based data mining methodology applied in this study. The experimental evaluation and results are then presented, and the paper concludes with a summary of key findings, limitations, and directions for future work.

II. RELATED WORK

The field of air quality prediction has undergone a significant paradigm shift, transitioning from deterministic physical models to data-driven machine learning approaches capable of capturing the stochastic and nonlinear nature of urban pollution dynamics Asnawi et al. [6]. As rapid industrialisation and urban expansion continue to degrade air quality globally, the ability to provide precise PM2.5 forecasts has become a critical requirement for public health authorities and smart city infrastructure Enciso-Díaz et al. [1], Ameer et al. [15]. This section reviews the progression of related research across four themes meteorological analysis, machine learning and ensemble methods, deep learning, and data mining frameworks critically evaluating each in relation to the objectives of this study.

A. Foundational Meteorological Analysis and Statistical Baselines

The cornerstone of air quality research lies in identifying the atmospheric drivers of pollutant dispersion. Early foundational work established that meteorological variables such as atmospheric pressure positively correlate with PM2.5 concentrations, while rainfall, wind speed, and humidity act as natural mitigators by dispersing or washing

out particulate matter Nakyai et al. [2]. Climate variability further affects pollutant dispersion through its influence on atmospheric circulation patterns, reinforcing the importance of including diverse meteorological inputs in predictive models Enciso-Díaz et al. [1], Vaduva et al. [3]. Unconventional data sources such as weather news have also been explored to capture a broader, decade-long perspective on how extreme climate phenomena intersect with air quality fluctuations Vaduva et al. [3]. These findings directly informed the feature selection strategy adopted in this study, where meteorological variables including temperature, humidity, wind speed, and pressure were included as primary predictors. However, while statistical approaches provide interpretable baselines, they consistently fail to capture the nonlinear interactions between environmental variables and pollutant concentrations Enciso-Díaz et al. [1]. Traditional linear models are frequently used as benchmarks but are critically limited by their inability to model high-dimensional variance, particularly in datasets with skewed distributions and multicollinearity challenges directly addressed through the preprocessing pipeline in this study Mandvi et al. [4].

B. Machine Learning and Ensemble Methods

To overcome the limitations of linear modelling, the literature has increasingly adopted ensemble machine learning algorithms. Research consistently shows that ensemble and tree-based machine learning methods outperform traditional statistical models when applied to air quality prediction, largely because these approaches are better equipped to learn the complex, nonlinear interactions that exist between environmental variables something that simpler regression techniques fundamentally struggle to capture Asnawi et al. [6]. Tree-based models, specifically Random Forest and XGBoost, have emerged as highly effective for predicting PM2.5 and AQI due to their high accuracy and lower error metrics compared to linear baselines Mandvi et al. [4], Renugadevi et al. [5]. A comparative performance analysis of multiple machine learning models for AQI prediction further validated that ensemble approaches achieve the highest predictive accuracy across diverse urban environments Mandvi et al. [4]. Random Forest has also been applied specifically in smart city environments, demonstrating robust performance when processing real-time environmental sensor data Renugadevi et al. [5], Ameer et al. [15]. XGBoost has shown particular promise for PM2.5 prediction, achieving strong correlation and low RMSE in daily concentration forecasting tasks, which directly supports its adoption as the primary model in this study Zhang et al. [8]. A significant challenge identified in recent studies is the class imbalance problem, where extreme pollution events occur far less frequently than moderate ones; researchers have addressed this by deploying boosting algorithms including AdaBoost, CatBoost, and LightGBM on meteorological monitoring data to improve sensitivity to high-pollution episodes Toharudin et al. [7]. While ensemble methods consistently deliver strong performance, many studies lack transparency in preprocessing decisions particularly multicollinearity handling and target transformation reducing reproducibility. This study addresses that gap explicitly through VIF-based feature selection and log transformation of the PM2.5 target.

C. Deep Learning Approaches and Their Limitations

The transition to deep learning has enabled modelling of complex temporal and spatial dependencies that traditional machine learning often overlooks. Attention-based deep learning models, such as the Integrated Multiple Directed

Attention framework, enhance temporal forecasting by prioritising relevant features across multiple time steps, outperforming standard recurrent networks for air pollution prediction Dairi et al. [10]. Hybrid CNN-LSTM frameworks have also gained traction, combining convolutional layers for spatial feature extraction with recurrent layers for temporal dependency modelling, achieving strong PM2.5 forecasting performance across multiple monitoring stations Li et al. [11]. Further hybrid architectures integrating multiple data modalities with deep learning have demonstrated additional improvements in forecasting accuracy Du et al. [12]. Advanced signal processing techniques such as wavelet decomposition have been combined with RBF-LSTM models to separate low-frequency regional pollution trends from high-frequency local variations before feeding data into deep learning architectures Chen et al. [14]. Spatiotemporal hybrid models incorporating atmospheric transport data have further pushed the boundaries of high-resolution PM2.5 prediction Feng et al. [13]. Despite these advances, deep learning models require large volumes of temporally dense data, significant computational resources, and sacrifice interpretability limiting their suitability for transparent and resource-efficient pipelines. Furthermore, AI and machine learning approaches for indoor air quality monitoring highlight that feature engineering and model selection remain critical even in deep learning settings Laton et al. [9]. This study deliberately adopts interpretable ensemble models over deep learning, prioritising reproducibility and cross-regional applicability on a global meteorological dataset.

D. Identified Gaps and Positioning of This Study

Despite the significant advances documented above, several important gaps remain across the literature. First, many studies focus on single-city or region-specific datasets, restricting the generalisability of findings across diverse urban and climatic environments Toharudin et al. [7], Feng et al. [13]. Second, the relative contribution of meteorological variables versus pollutant co-features to PM2.5 prediction is rarely isolated and quantified through controlled experimentation Asnawi et al. [6]. Third, preprocessing decisions including outlier treatment, target variable transformation, and multicollinearity resolution are frequently underreported across both machine learning and deep learning studies, limiting reproducibility Mandvi et al [4], Laton et al. [9]. This study directly addresses all three gaps by applying a rigorous CRISP-DM-based pipeline to a large-scale global meteorological dataset spanning over 120,000 records, conducting a three-tier ablation study to isolate the contribution of raw meteorological features, engineered features, and pollutant co-variables respectively, and providing a fully documented and reproducible preprocessing and modelling workflow.

TABLE I: Summary of Related Work

Ref	Method	Dataset Scope	Key Limitation
[1]	Statistical review	Global, multi-city	No predictive model built
[2]	Pearson correlation, regression	Single region, Thailand	Limited to linear relationships

[3]	Weather data processing	Decade-long, news + weather	No ML prediction model
[4]	RF, SVM, ANN	Single city, Gorakhpur	Region-specific, limited generalisability
[5]	Random Forest	Urban environment	No feature engineering discussed
[6]	Systematic literature review	Multi-study review	No original experimental contribution
[7]	XGBoost, AdaBoost, CatBoost, LightGBM	Jakarta, Indonesia	Single city, classification only
[8]	XGBoost	Single city, China	No preprocessing pipeline reported
[9]	AI/ML, feature engineering	Indoor environments	Limited to indoor air quality
[10]	Attention-based deep learning	Multi-station	High computational cost, low interpretability
[11]	CNN-LSTM hybrid	Single city	Requires dense temporal data
[12]	Hybrid deep learning	Multi-source	Complex architecture, limited reproducibility
[13]	Spatiotemporal hybrid model	Regional, China	Region-specific, high complexity
[14]	Wavelet + RBF-LSTM	Single city, Taiwan	No cross-regional validation
[15]	Comparative ML	Smart city environments	Does not isolate feature contributions

In conclusion, the literature demonstrates a clear evolution from traditional statistical baselines to advanced ensemble and hybrid deep learning techniques for PM2.5 prediction. While ensemble methods such as Random Forest and XGBoost have shown strong predictive, existing studies remain limited by single-city datasets, underreported preprocessing pipelines, and insufficient isolation of feature contributions. Deep learning approaches, despite their accuracy, sacrifice interpretability and cross-regional generalisability. This study addresses these gaps by applying a structured CRISP-DM-based pipeline to a global meteorological dataset, with rigorous feature engineering, VIF-based multicollinearity resolution, and a three-tier ablation study to quantify the contribution of meteorological, engineered, and pollutant features to PM2.5 prediction.

III. METHODOLOGY

This study follows the CRISP-DM framework to structure the end-to-end analytical pipeline. Of the six standard CRISP-DM phases, five are directly addressed in this study: business

understanding, data understanding, data preparation, modelling, and evaluation. The deployment phase is partially addressed through the export and saving of all trained models for reproducibility purposes. Each phase is described below with its rationale in the context of PM2.5 prediction.

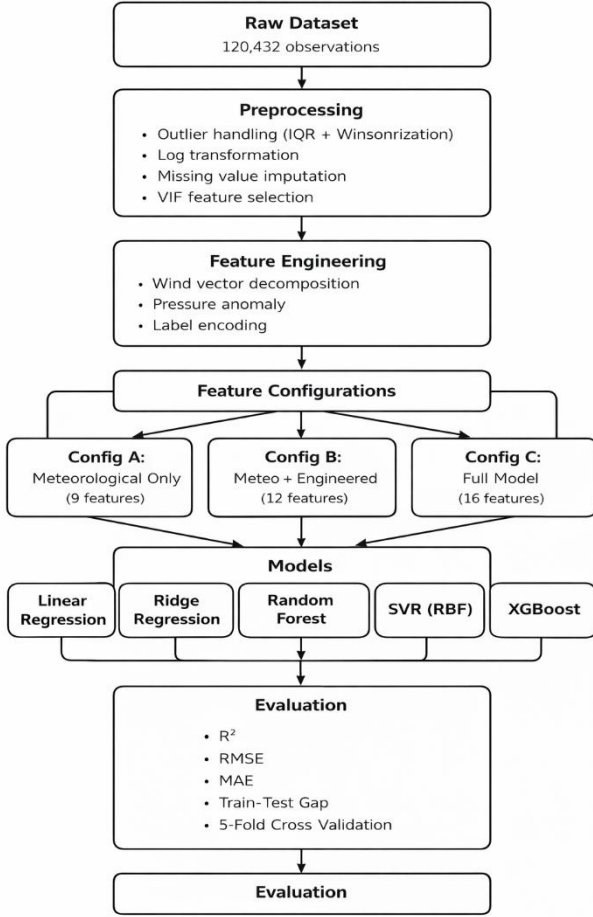


Fig. 1. Overview of the CRISP-DM-based PM2.5 prediction pipeline.

A. Dataset Description

The primary objective of this study is to predict continuous PM2.5 concentrations from meteorological and environmental features, directly addressing RQ1 and RQ2. PM2.5 is selected as the target variable in preference to the categorical Air Quality Index, as it preserves predictive resolution and enables regression-based modelling [4]. The dataset used is the Global Weather Repository, sourced from Kaggle, comprising 120,432 observations and 41 columns aggregated from weather stations across multiple countries worldwide [16]. Features are organised into three groups: twelve raw meteorological measurements including temperature, humidity, wind speed, pressure, and precipitation; four pollutant co-features representing Carbon Monoxide, Ozone, Nitrogen Dioxide, and Sulphur Dioxide; and one categorical weather condition descriptor [2]. PM10 is intentionally excluded despite its strong correlation with PM2.5 (Pearson $r \approx 0.65$), as both pollutants share common emission sources and including PM10 would inflate model performance through spurious correlation rather than genuine meteorological signal [18]. Initial exploratory analysis reveals that PM2.5 is heavily right-skewed, with a mean of approximately $35 \mu\text{g}/\text{m}^3$ and extreme values exceeding $500 \mu\text{g}/\text{m}^3$, and that over 55% of observations

exceed the WHO recommended safe threshold of $15 \mu\text{g}/\text{m}^3$.

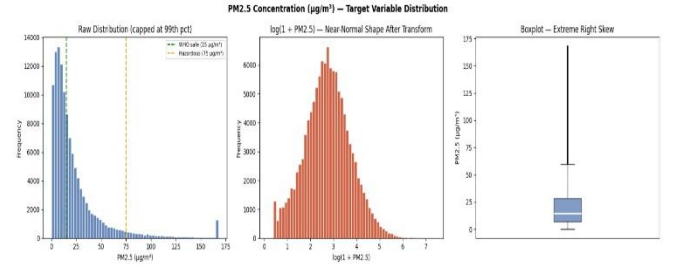


Fig. 2. PM2.5 concentration distribution: raw histogram capped at 99th percentile (left), $\log(1 + \text{PM2.5})$ near-normal distribution after transformation (centre), and boxplot showing extreme right skew (right).

B. Data Understanding and Preprocessing

A rigorous multi-step preprocessing pipeline is applied to ensure data quality and model reliability. Before any analysis began, values that were physically impossible were flagged and replaced with NaN. This included negative pollutant concentrations, wind speeds above 300 km/h, and atmospheric pressure readings exceeding 1,100 mb none of which can realistically occur in natural atmospheric conditions, and which were therefore treated as sensor errors rather than genuine observations. Second, redundant columns are removed based on three criteria: unit-duplicate features such as temperature in Fahrenheit are dropped as they are perfect linear transformations of their metric counterparts; astronomical variables such as moon phase and moonrise are excluded as they have no physical link to PM2.5; and derived AQI index columns are removed to prevent target leakage. Third, univariate outlier detection is performed using the Interquartile Range method. For the target variable, winsorisation at the 99th percentile is applied rather than row deletion, preserving sample size while controlling the influence of extreme values. Multivariate outlier detection using Mahalanobis distance is also conducted, with flagged observations retained as they may represent genuine high-pollution events [19]. Fourth, a log transformation is applied to the PM2.5 target using $\log(1 + \text{PM2.5})$, reducing skewness from approximately 2.5 to 0.1 and producing a near-normal distribution that satisfies linear model assumptions. All models are trained and evaluated in log-space. Fifth, missing values fewer than 0.01% of rows are imputed using median imputation fitted exclusively on the training set to prevent data leakage. Finally, the dataset is split 80/20 into training and test sets, with feature scaling applied for distance-based and regularised models only, as tree-based models are scale-invariant.

C. Feature Engineering

Four engineered features are constructed to capture domain-specific physical relationships not represented by the raw measurements. Wind direction is tricky to work with because it wraps around 359° and 1° are actually next to each other on a compass, but numerically they are 358 units apart, which would confuse any standard model trying to learn from that feature. This is resolved by decomposing wind speed and direction into east-west and north-south Cartesian components, encoding both direction and magnitude in a format suitable for all model types. A heat stress index is constructed as the product of temperature and relative humidity, capturing the combined thermodynamic effect of heat and moisture on atmospheric conditions associated with

thermal inversions that trap pollutants near the surface [20]. A pressure anomaly feature is derived as the absolute deviation from standard sea-level pressure, as deviations indicate the presence of high- or low-pressure systems that directly affect vertical mixing and pollutant dispersion. Finally, the categorical weather condition variable is label-encoded to enable its use as a model input.

Following feature construction, Variance Inflation Factor analysis is performed iteratively to remove multicollinear features, with pollutant features protected from removal given their domain importance [9] [23]. Features exceeding a VIF threshold of 5 are progressively eliminated until all remaining features satisfy the threshold, resulting in the removal of temperature, feels-like temperature, and the heat stress index due to high collinearity. The final feature set is further validated using Lasso-based embedded selection and Mutual Information scoring [8], confirming that all retained features carry meaningful predictive signal for PM2.5.

D. Model Development

Five regression models are developed spanning linear, regularised, kernel-based, and ensemble-based approaches. Linear Regression serves as the baseline, establishing a performance floor against which all other models are compared [24]. Ridge Regression builds on the linear baseline by applying L2 regularisation, which penalises large coefficients and reduces the model's sensitivity to correlated features, with the optimal regularisation strength determined through five-fold cross-validation. Random Forest takes a fundamentally different approach, constructing a large ensemble of decorrelated decision trees by training each tree on a different random subset of the data and features; key hyperparameters including tree depth, minimum leaf size, and bootstrap fraction were tuned via grid search to strike the right balance between bias and variance [5]. XGBoost builds trees sequentially, with each tree correcting the residuals of its predecessors, and applies four regularisation mechanisms L1 penalty, L2 penalty, column subsampling, and row subsampling making it the most richly regularised model in the comparison [8]. Support Vector Regression with a radial basis function kernel captures nonlinear feature relationships and is trained on a representative subsample due to its computational complexity scaling with dataset size [25].

To directly address RQ2, all four primary models are additionally trained under two restricted feature configurations: a raw meteorological only setting using weather measurements alone, and a raw plus engineered meteorological setting adding the interaction features but excluding pollutant co-variables. This three-tier ablation design isolates the contribution of each feature group to PM2.5 prediction accuracy, enabling a controlled quantitative comparison between meteorological-only and full-feature model performance.

E. Evaluation Framework

All models are evaluated consistently using R^2 , RMSE, and MAE computed in log-space to ensure direct comparability. The train-test R^2 gap is monitored as an overfitting diagnostic. AIC and BIC are computed for linear models to assess the trade-off between goodness of fit and model complexity. Five-fold cross-validation provides robust generalisation estimates independent of the single train-test split, and a paired t-test on squared errors formally tests

whether performance differences between the top models are statistically significant rather than arising by chance. The complete code and dataset can be accessed at: <https://drive.google.com/drive/folders/1pt6Zctfmo6C9gCbbkAFyZVkvzES9tqLGm?usp=sharing>

IV. RESULT AND DISCUSSIONS

This section presents the experimental results across three feature configurations full model, raw meteorological only, and raw plus engineered meteorological to systematically evaluate the contribution of different feature groups and model classes to PM2.5 prediction. All metrics are computed in $\log(1 + \text{PM2.5})$ space to ensure consistent comparison, with performance assessed using R^2 , RMSE, MAE, and five-fold cross-validation. Results are then discussed in relation to the two research questions guiding this study.

A. Full Model Performance

Five regression models are evaluated on the held-out test set comprising 24,087 observations, with all metrics computed in $\log(1 + \text{PM2.5})$ space to ensure direct comparability across model classes. XGBoost emerges as the best-performing model overall, achieving a test R^2 of 0.7749 and a test RMSE of 0.4648, with a train-test gap of just 0.0232 well within the acceptable threshold confirming that the regularisation strategy successfully prevented overfitting across 300 sequential trees [8]. The cross-validation results further reinforce this, with XGBoost achieving a mean R^2 of 0.7982 ± 0.0017 across five folds, the tightest standard deviation of any model in the study, confirming both strong and stable generalisation. SVR with an RBF kernel achieves the second highest test R^2 of 0.6993 with a train-test gap of 0.0172, demonstrating that kernel-based nonlinear modelling is effective on this dataset despite being trained on only a 20,000-row subsample due to computational constraints. Random Forest achieves a test R^2 of 0.6566 with the smallest train-test gap of all models (0.0125), reflecting the strong variance-reduction properties of ensemble averaging across 200 decorrelated trees, and achieves a cross-validation mean R^2 of 0.6681 ± 0.0041 . Linear Regression and Ridge Regression both plateau at a test R^2 of 0.4183 with nearly identical RMSE of 0.7473 and cross-validation RMSE of 0.7469 ± 0.0038 , confirming that the VIF-based feature selection applied during preprocessing had already resolved most multicollinearity, leaving Ridge's L2 penalty with little room to improve beyond the OLS baseline. The consistent performance hierarchy across all metrics XGBoost > SVR > Random Forest > Ridge $\approx$ Linear establishes that ensemble-based nonlinear models substantially outperform linear approaches on this task, directly answering RQ1 [6], [4].

TABLE II: Full Model Performance Comparison (All metrics in log-space)

Model	Train R^2	Test R^2	Gap	Test RMSE	Test MAE
Linear Regression	0.4124	0.4183	-0.0059	0.7473	0.5893
Ridge Regression	0.4124	0.4183	-0.0059	0.7473	0.5893

Random Forest	0.6691	0.6566	0.0125	0.5742	0.4501
SVR (RBF Kernel)	0.7165	0.6993	0.0172	0.5373	0.4104
XGBoost	0.7981	0.7749	0.0232	0.4648	0.3583

which are captured indirectly through co-pollutant measurements.

TABLE III: Meteorological-Only Model Results (log-space)

Model	Train R ²	Test R ²	R ² Gap	RMSE	MAE
XGBoost (Raw+Eng)	0.4568	0.4477	0.0091	0.9336	0.7312
XGBoost (Raw Only)	0.4432	0.4341	0.0091	0.9448	0.7407
RF (Raw+Eng)	0.4101	0.4009	0.0092	0.9724	0.7610
RF (Raw Only)	0.3987	0.3920	0.0067	0.9798	0.7678
Ridge (Raw+Eng)	0.3037	0.3020	0.0017	1.0495	0.8178
Linear (Raw Only)	0.2926	0.2910	0.0016	1.0576	0.8247

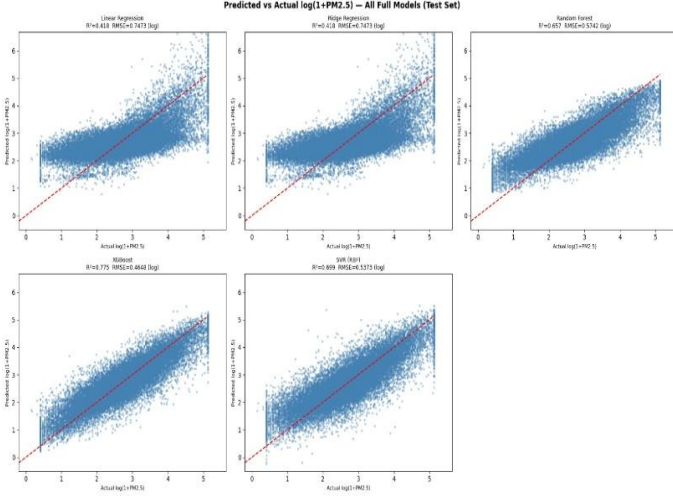


Fig. 3. Predicted vs Actual (All Models)

A paired t-test comparing the squared prediction errors of XGBoost and SVR on the test set yields $t = -29.79$ and $p < 0.0001$, conclusively rejecting the null hypothesis of no difference. The mean squared error of XGBoost (0.2161) is substantially lower than that of SVR (0.2887), confirming that XGBoost's superior performance is statistically significant and not attributable to random variation in the test partition.

B. Contribution of Feature Groups

To directly address RQ2, all models are retrained under two restricted feature configurations: raw meteorological measurements only, and raw plus engineered meteorological features excluding pollutant co-variables. This three-tier design isolates the contribution of each feature group to predictive accuracy. When trained on raw meteorological features alone, the best-performing model is XGBoost with a test R² of 0.4000, while Random Forest achieves only 0.2577 and Ridge and Linear models both reach 0.1483.

Adding the engineered features—wind vector decomposition and pressure anomaly—produces a modest but consistent improvement for most model types: XGBoost gains 0.0037 R² points (reaching 0.4037), and Ridge and Linear both gain 0.0155 (reaching 0.1638), although Random Forest shows a marginal decline of -0.0044 , likely due to the additional feature interactions introduced by the engineering step interacting unpredictably with the bootstrap sampling mechanism under the restricted feature set. The pollutant co-features (Carbon Monoxide, Ozone, Nitrogen Dioxide, and Sulphur Dioxide) account by far for the largest performance gain. XGBoost gains 0.3712 R² points when moving from the engineered meteorological configuration to the full model, while Ridge and Linear both gain 0.2545. These findings directly answer RQ2: meteorological variables alone explain only approximately 40% of PM2.5 variance at best, and atmospheric conditions govern pollutant dispersion and transport but not the rate at which pollutants are emitted [1], [2]. The remaining variance is driven by emission-source factors such as industrial activity, traffic, and urban land use,

C. Feature Importance and SHAP Analysis

Feature importance analysis and SHAP explainability provide complementary perspectives on the drivers of PM2.5 prediction. Mutual Information scoring reveals that Sulphur Dioxide (MI = 0.602), Nitrogen Dioxide (MI = 0.458), and Carbon Monoxide (MI = 0.399) are the three most informative features overall, reflecting strong nonlinear dependencies with PM2.5 that Pearson correlation alone would underestimate. Among meteorological features, UV index (MI = 0.288) and humidity (MI = 0.085) carry the most predictive signal. SHAP analysis of the best-performing XGBoost model confirms Carbon Monoxide as the single most influential predictor in terms of prediction impact magnitude, followed by humidity and Nitrogen Dioxide [7]. The SHAP dependence plot for humidity reveals a clear nonlinear pattern: moderate humidity between 40% and 70% has minimal influence on PM2.5, while very high humidity above 80% is associated with elevated concentrations particularly at higher temperatures, reflecting the physical phenomenon of hygroscopic growth where particulate matter absorbs atmospheric moisture and increases in measured mass [2]. OLS coefficient analysis confirms that temperature ($t = 32.44$, $p < 0.001$), visibility ($t = -47.36$, $p < 0.001$), humidity ($t = -22.35$, $p < 0.001$), and precipitation ($t = -8.95$, $p < 0.001$) are all statistically significant linear predictors of log-PM2.5, while atmospheric pressure does not reach significance at the 5% level ($p = 0.060$).

D. Model Diagnostics

Residual diagnostics applied to the best-performing XGBoost model confirm that predictions are unbiased across the fitted range, with no systematic over- or under-prediction visible in the residuals-versus-fitted plot. The Shapiro-Wilk test confirms that residuals are not normally distributed ($W = 0.9942$, $p < 0.001$), which is expected given the heavy-tailed nature of PM2.5 data even after log transformation. The Breusch-Pagan test confirms heteroscedasticity across all models ($p < 0.001$ in every case), indicating that residual variance is not constant—a violation of OLS assumptions that further motivates the adoption of tree-based models which make no homoscedasticity assumption [9]. Looking at the learning curves, XGBoost's validation RMSE kept dropping as more training data was added, which suggests the model had not yet reached its

performance ceiling and would likely improve further if more observations were available beyond the 120,432 records used here.

E. Discussion

The results collectively provide clear quantitative answers to both research questions. For RQ1, XGBoost demonstrates that machine learning can predict PM2.5 with substantial accuracy test R^2 of 0.7749 and cross-validation R^2 of 0.7982 ± 0.0017 when meteorological and pollutant features are combined, with ensemble-based nonlinear models consistently and significantly outperforming linear approaches. For RQ2, the ablation study confirms that meteorological variables alone explain at most 40% of PM2.5 variance, with pollutant co-features accounting for an additional 37 percentage points of explained variance in the best model. This finding underscores that emission-source factors play a dominant and irreplaceable role that atmospheric conditions alone cannot capture. The fact that Linear Regression and Ridge Regression performed almost identically speaks to how well the preprocessing pipeline handled multicollinearity before modelling even began. Meanwhile, the same performance ranking held up across both cross-validation and the held-out test set, giving confidence that the results are genuine and not a product of any particular data split.

V. CONCLUSION & FUTURE WORK

This study presented a structured CRISP-DM-based data mining pipeline for predicting PM2.5 concentrations from a large-scale global meteorological dataset comprising over 120,000 observations. In total, five machine learning models were built and tested across three different feature configurations, with each experiment producing consistent and reproducible results.

Addressing RQ1, the results demonstrate that machine learning models can predict PM2.5 with substantial accuracy when meteorological and pollutant features are combined, with XGBoost achieving the strongest performance at a test R^2 of 0.7749 and a cross-validation R^2 of 0.7982 ± 0.0017 . Ensemble-based nonlinear models consistently and significantly outperformed linear baselines, with the paired t-test confirming XGBoost's superiority over SVR at $p < 0.0001$. Addressing RQ2, the three-tier ablation study reveals that meteorological variables alone explain at most 40% of PM2.5 variance, with pollutant co-features accounting for an additional 37 percentage points of explained variance. This finding underscores that atmospheric conditions govern pollutant dispersion but not emission rates, and that factors such as industrial activity, traffic, and urban land use play a dominant and irreplaceable role in determining PM2.5 levels.

The study makes three contributions to knowledge. First, it provides a fully documented and reproducible preprocessing pipeline incorporating sensor error correction, IQR-based outlier treatment, log transformation, VIF-based multicollinearity removal, and median imputation that can be directly applied to other global environmental datasets. Second, it quantifies through controlled experimentation the relative contribution of meteorological features, engineered features, and pollutant co-variables to PM2.5 prediction accuracy. Third, it demonstrates that interpretable ensemble

models applied to globally diverse data can achieve strong predictive performance without the computational cost or opacity of deep learning architectures.

Several limitations of this study should be acknowledged. The dataset is a static snapshot and does not capture temporal trends or seasonal dynamics, which may limit the model's applicability to real-time forecasting scenarios. SVR was trained on a 20,000-row subsample due to computational constraints, which may have modestly affected its performance relative to a full-data training run. Furthermore, the learning curve analysis indicates that XGBoost's validation performance had not yet plateaued at the maximum training size, suggesting that additional data would likely improve predictive accuracy further.

Given more time, several extensions would strengthen this work. Incorporating temporal features such as hour of day, day of week, and seasonal indicators would enable the model to capture diurnal and seasonal pollution cycles not represented in the current feature set. Including more years of data and covering a wider range of geographic locations would help the model perform more reliably across different regions and climates. Finally, applying SHAP-guided feature selection to iteratively refine the feature set, and exploring lightweight gradient boosting alternatives such as LightGBM for more efficient deployment, represent promising directions for future work.

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